

Dermatological Disorders: Drug-induced Skin Reactions

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Introduction

- Drug-induced skin reactions can be irritant or allergic in origin.
- Allergic drug reactions are classified into exanthematous, urticarial, blistering, and pustular eruptions.
- Severe cutaneous adverse reactions to drugs (SCARs) include Stevens–Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), and drug reaction with eosinophilia and systemic symptoms (DRESS).

Pathophysiology

- **Exanthematous drug reactions** include maculopapular rashes and drug hypersensitivity syndrome
- **Urticarial reactions** include urticaria, angioedema, and serum sicknesslike reactions.
- **Blistering reactions** include fixed drug eruptions, Stevens–Johnson syndrome, and toxic epidermal necrolysis
- **Pustular eruptions** include acneiform drug reactions and acute generalized exanthematous pustulosis (AGEP)

- **Drug-induced hyperpigmentation** may be related to increased melanin (eg, hydantoins), direct deposition (eg, silver, mercury, tetracyclines, and antimalarials), or other mechanisms (eg, fluorouracil).
- **Drug-induced photosensitivity reactions** may be **phototoxic** (a nonimmunologic reaction) or photoallergic (an immunologic reaction).
 - ✓ Medications associated with phototoxicity include amiodarone, tetracyclines, sulfonamides, psoralens, and coal tar.
- Common causes of **photoallergic reactions** include:
 - ✓ sulfonamides, sulfonylureas, thiazides, NSAIDs, chloroquine, and carbamazepine

Clinical Presentation

- Maculopapular skin reaction presents with erythematous macules and papules that may be pruritic.
 - Lesions usually begin within 7–10 days after starting the offending medication and generally resolve within 7–14 days after drug discontinuation.
 - Commonly by penicillins, cephalosporins, sulfonamides, and some anticonvulsants.
 - Antibiotics may be d/c before reaction occurs



Drug Hypersensitivity Syndrome

- also known as *drug reaction with eosinophilia and systemic symptoms* or **DRESS**) is an **exanthematous eruption** accompanied by fever, lymphadenopathy, and multiorgan involvement (kidneys, liver, lung, bone marrow, heart, and brain).
- Signs and symptoms begin 1–4 weeks after starting the offending drug, and the reaction may be fatal if not promptly treated.
- Drugs implicated include allopurinol, sulfonamides, some anticonvulsants (barbiturates, phenytoin, carbamazepine, and lamotrigine), and dapsone.



Table 1. RegiSCAR drug reaction with eosinophilia and systemic symptoms (DRESS) inclusion criteria *** [11].

1	Hospitalisation
2	Skin eruption
3	Fever $>38^{\circ}\text{C}$
4	Lymphadenopathy at least 2 sites
5	Involvement of at least one internal organ
6a	Lymphocytosis ($>4 \times 10^3/\mu\text{L}$) or lymphopenia ($<1.5 \times 10^3/\mu\text{L}$)
6b	Eosinophilia $>10\%$ or $>700/\mu\text{L}$
6c	Thrombocytopaenia ($<120 \times 10^3/\mu\text{L}$)

*** At least 3 criteria must be fulfilled.

Urticaria and angioedema

- are simple eruptions that are caused by drugs in 5%–10% of cases.
- Other causes are foods (most common) and physical factors such as cold or pressure, infections, and latex exposure.
- Urticaria may be the first sign of an emerging anaphylactic reaction characterized by hives, extremely pruritic red raised wheals, angioedema, and mucous membrane swelling that typically occurs within minutes to hours.
- Offending drugs include penicillins and related antibiotics, aspirin, sulfonamides, radiograph contrast media, and opioids.
- Serum sickness-like reactions are complex urticarial eruptions presenting with fever, rash (usually urticarial), and arthralgias usually within 1–3 weeks after starting the offending drug.



Fixed drug eruptions

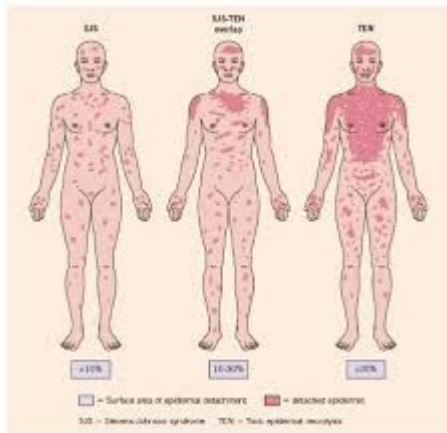
- present as pruritic, red, raised lesions that may blister.
- Symptoms can include burning or stinging. Lesions may evolve into plaques. These so-called fixed eruptions recur in the same area each time the offending drug is given.
- Lesions appear and disappear within minutes to days, leaving hyperpigmented skin for months.
- Usual offenders include *tetracyclines*, *barbiturates*, *sulfonamides*, *codeine*, *phenolphthalein*, and *NSAIDs*



Stevens–Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN)



- are blistering eruptions that are rare but severe and life-threatening.
- Onset occurs within 7–14 days after drug exposure.
- Pts present with generalized tender/ painful bullous formation with fever, headache, and respiratory symptoms leading to rapid clinical deterioration.
- Lesions show rapid confluence and spread, resulting in extensive epidermal detachment and sloughing. This may result in marked fluid loss, hypotension, electrolyte imbalances, and secondary infections.
- Usual offending drugs include sulfonamides, penicillins, some anticonvulsants (hydantoins, carbamazepine, barbiturates, and lamotrigine), NSAIDs, and allopurinol.



- **Acneiform drug reactions** are pustular eruptions that induce acne. Onset is within 1–3 weeks.
- Common culprits include corticosteroids, androgenic hormones, some anticonvulsants, isoniazid, and lithium.
- **Acute generalized exanthematous pustulose (AGEP)** has an **acute onset** (within days after starting the offending drug), fever, diffuse erythema, and many pustules.
- Generalized desquamation occurs 2 weeks later.
- Common offending drugs include ***β-lactam antibiotics, macrolides, and calcium channel blockers.***
- **Sun-induced skin reactions** appear similar to a sunburn and present with erythema, papules, edema, and sometimes vesicles.
- They appear in areas exposed to sunlight (eg, ears, nose, cheeks, forearms, and hands).

DIAGNOSIS

- A comprehensive patient history is important to obtain the following information:
 - ✓ **Signs and symptoms** (onset, progression, timeframe, lesion location and description, presenting symptoms, and previous occurrence)
 - ✓ Urgency (severity, area, and extent of skin involvement; signs of a systemic/ generalized reaction or disease condition)
 - ✓ Medication history (temporal correlation, previous exposure, and nonprescribed products)
 - ✓ Differential diagnosis
- Lesion assessment includes identifying macules, papules, nodules, blisters, plaques, and lichenification.
- Some skin conditions cause more than one type of lesions.
- Inspect lesions for color, texture, size, and temperature. Areas that are oozing, erythematous, and warm to the touch may be infected.

Treatment

- **Goals of Treatment:**
 - Relieve bothersome symptoms,
 - remove precipitating factors,
 - prevent recurrences,
 - avoid adverse treatment effects, and
 - improve quality of life.

Treatment

- If a drug-induced skin reaction is suspected, the most important treatment is discontinuing the suspected drug as quickly as possible and avoiding use of potential cross-sensitizers.
- The next step is to control symptoms (eg, pruritus).
- Signs or symptoms of a systemic or generalized reaction may require additional supportive therapy.
- For **high fevers**, acetaminophen is more appropriate than aspirin or another NSAID, which may exacerbate some skin lesions

- Most maculopapular reactions disappear within a few days after discontinuing the agent, so symptomatic control of the affected area is the primary intervention.
 - Topical corticosteroids and oral antihistamines can relieve pruritus. In severe cases, a short course of systemic corticosteroids may be warranted.
- Treatment of fixed drug reactions involves removal of the offending agent.
- Other therapeutic measures include topical corticosteroids, oral antihistamines to relieve itching, and perhaps cool water compresses on the affected area.

- Photosensitivity reactions typically resolve with drug discontinuation.
- Some patients benefit from topical corticosteroids and oral antihistamines, but these are relatively ineffective.
- Systemic corticosteroids (eg, oral prednisone 1 mg/kg/day tapered over 3 weeks) are more effective.

- **For life-threatening SJS/TEN,**
 - Supportive measures such as maintenance of adequate blood pressure, fluid and electrolyte balance, broad-spectrum antibiotics and vancomycin for Secondary infections, **and IV immunoglobulin (IVIG)** may be appropriate.
 - Corticosteroid use** is controversial; if used, employ relatively high doses initially, followed by rapid tapering as soon as disease progression stops.
- **Inform patients** about the suspected drug, potential drugs to avoid in the future, and which drugs may be used instead.
- Give pts with photosensitivity reactions information about preventive measures, such as use of sunscreens and sun avoidance.

EVALUATION OF THERAPEUTIC OUTCOMES

- Provide patients with information regarding causative factors, avoidance of substances that trigger skin reactions, and potential benefits and limitations on nondrug and drug therapy